11.13. DIFFERENTIAL EQUATIONS OF THE FIRST ORDER AND HIGHER DEGREE

So far, we have discussed differential equations of the first order and first degree. Now we shall study differential equations of the first order and degree higher than the first. For convenience, we denote $\frac{dy}{dx}$ by p.

A differential equation of the first order and nth degree is of the form

$$p^n + P_1 p^{n-1} + P_2 p^{n-2} + \ldots\ldots + P_n = 0 \qquad \ldots(1)$$

where $P_1, P_2, \ldots\ldots, P_n$ are functions of x and y.

Since it is a differential equation of the first order, its general solution will contain only one arbitrary constant.

In the various cases which follow, the problem is reduced to that of solving one or more equations of the first order and first degree.

11.14. EQUATIONS SOLVABLE FOR p

Resolving the left hand side of (1) into n linear factors, we have

$$[p - f_1(x, y)]\,[p - f_2(x, y)], \ldots\ldots, [p - f_n(x, y)] = 0$$

which is equivalent to $\quad p - f_1(x, y) = 0, p - f_2(x, y) = 0, \ldots\ldots, p - f_n(x, y) = 0$

Each of these equations is of the first order and first degree and can be solved by the methods already discussed.

If the solutions of the above n component equations are

$$F_1(x, y, c) = 0, F_2(x, y, c) = 0, \ldots\ldots, F_n(x, y, c) = 0$$

then the general solution of (1) is given by $F_1(x, y, c).\ F_2(x, y, c) \ldots\ldots F_n(x, y, c) = 0$.

Example 1. *Solve* $x^2\left(\dfrac{dy}{dx}\right)^2 + xy\dfrac{dy}{dx} - 6y^2 = 0.$

Sol. The given equation is $x^2p^2 + xyp - 6y^2 = 0$ where $p = \dfrac{dy}{dx}$

Factorising $\quad (xp + 3y)(xp - 2y) = 0 \Rightarrow xp + 3y = 0$ or $xp - 2y = 0$

Now $\quad xp + 3y = 0 \Rightarrow x\dfrac{dy}{dx} + 3y = 0$ or $\dfrac{dy}{y} + 3\dfrac{dx}{x} = 0$

Integrating $\quad \log y + 3\log x = \log c$ or $x^3y = c$

Also $\quad xp - 2y = 0 \Rightarrow x\dfrac{dy}{dx} - 2y = 0$ or $\dfrac{dy}{y} - 2\dfrac{dx}{x} = 0$

Integrating $\quad \log y - 2\log x = \log c$ or $\dfrac{y}{x^2} = c$ or $y = cx^2$

$\therefore$ The general solution of the given equation is $(x^3y - c)(y - cx^2) = 0$.

Example 2. *Solve* $xyp^2 + p(3x^2 - 2y^2) - 6xy = 0.$

Sol. Solving the given equation for p, we have

$$p = \frac{-(3x^2 - 2y^2) \pm \sqrt{(3x^2 - 2y^2)^2 + 24x^2y^2}}{2xy}$$

$$= \frac{(2y^2 - 3x^2) \pm (3x^2 + 2y^2)}{2xy} = \frac{2y}{x} \text{ or } -\frac{3x}{y}$$

Now $\quad p = \dfrac{2y}{x} \Rightarrow \dfrac{dy}{dx} = \dfrac{2y}{x}$ or $\dfrac{dy}{y} - \dfrac{2dx}{x} = 0$

Integrating $\log y - 2\log x = \log c$ or $\dfrac{y}{x^2} = c$ or $y = cx^2$

Also $\quad p = -\dfrac{3x}{y} \Rightarrow \dfrac{dy}{dx} = -\dfrac{3x}{y}$ or $ydy + 3xdx = 0$

Integrating $\quad \dfrac{y^2}{2} + \dfrac{3x^2}{2} = C$ or $y^2 + 3x^2 = c$

$\therefore$ The general solution of the given equation is $(y - cx^2)(y^2 + 3x^2 - c) = 0$.

Example 3. *Solve* $p^2 + 2py \cot x = y^2$.

Sol. The given equation can be written as $(p + y \cot x)^2 = y^2 (1 + \cot^2 x)$

or $$p + y \cot x = \pm\, y \operatorname{cosec} x$$

$\therefore$ The component equations are $p = y(-\cot x + \operatorname{cosec} x)$

and $$p = y(-\cot x - \operatorname{cosec} x)$$

From (1), $\dfrac{dy}{dx} = y(-\cot x + \operatorname{cosec} x)$ or $\dfrac{dy}{y} = (-\cot x + \operatorname{cosec} x)\, dx$

Integrating $\log y = -\log \sin x + \log \tan \dfrac{x}{2} + \log c = \log \dfrac{c \tan \frac{x}{2}}{\sin x}$

or $$y = \frac{c \tan \frac{x}{2}}{2 \sin \frac{x}{2} \cos \frac{x}{2}} = \frac{c}{2 \cos^2 \frac{x}{2}} = \frac{c}{1 + \cos x}$$

or $$y(1 + \cos x) = c$$

From (2), $\dfrac{dy}{dx} = y(-\cot x - \operatorname{cosec} x)$ or $\dfrac{dy}{y} = (-\cot x - \operatorname{cosec} x)\, dx$

Integrating $\log y = -\log \sin x - \log \tan \dfrac{x}{2} + \log c = \log \dfrac{c}{\sin x \tan \frac{x}{2}}$

or $$y = \frac{c}{2 \sin^2 \frac{x}{2}} = \frac{c}{1 - \cos x} \quad \text{or} \quad y(1 - \cos x) = c$$

$\therefore$ The general solution of the given equation is $y(1 \pm \cos x) = c$.

11.15. EQUATIONS SOLVABLE FOR y

If the equation is solvable for y, we can express y explicitly in terms of x and p. Thus, the equations of this type can be put as $y = f(x, p)$ ...(1)

Differentiating (1) w.r.t. x, we get $\frac{dy}{dx} = p = F\left(x, p, \frac{dp}{dx}\right)$ ...(2)

Equation (2) is a differential equation of first order in p and x.

Suppose the solution of (2) is $\phi(x, p, c) = 0$ ...(3)

Now elimination of p from (1) and (3) gives the required solution.

If p cannot be easily eliminated, then we solve equations (1) and (3) for x and y to get

$$x = \phi_1(p, c),\ y = \phi_2(p, c)$$

These two relations together constitute the solution of the given equation with p as parameter.

Example 1. *Solve* $y + px = x^4p^2$. (Madras, 1996)

Sol. Given equation is $\quad y = -px + x^4p^2$ ...(1)

Differentiating both sides w.r.t. x,

$$\frac{dy}{dx} = p = -p - x\frac{dp}{dx} + 4x^3p^2 + 2x^4p\frac{dp}{dx}$$

or $$2p + x\frac{dp}{dx} - 2px^3\left(2p + x\frac{dp}{dx}\right) = 0 \quad \text{or} \quad \left(2p + x\frac{dp}{dx}\right)(1 - 2px^3) = 0$$

Discarding the factor $(1 - 2px^3)$, we have $2p + x\frac{dp}{dx} = 0$ or $\frac{dp}{p} + 2\frac{dx}{x} = 0$

Integrating $\quad \log p + 2\log x = \log c \quad$ or $\quad \log px^2 = \log c \quad$ or $\quad px^2 = c$

or $$p = \frac{c}{x^2}.$$

Putting this value of p in (1), we have $y = -\frac{c}{x} + c^2$ which is the required solution.

Example 2. *Solve* $y = 2px - p^2$.

Sol. The given equation is $y = 2px - p^2$ ...(1)

Differentiating both sides w.r.t. x, $\frac{dy}{dx} = p = 2p + 2x\frac{dp}{dx} - 2p\frac{dp}{dx}$

or $$p + (2x - 2p)\frac{dp}{dx} = 0 \quad \text{or} \quad p\frac{dx}{dp} + 2x - 2p = 0$$

or $$\frac{dx}{dp} + \frac{2}{p}x = 2$$

which is a linear equation.

$$\text{I.F.} = e^{\int \frac{2}{p}dp} = e^{2\log p} = p^2$$

$\therefore$ The solution of (2) is $\quad x\,(\text{I.F.}) = \int 2\,(\text{I.F.})\,dp + c \quad$ or $\quad xp^2 = \int 2p^2 dp + c$

or $$xp^2 = \frac{2}{3}p^3 + c \quad \text{or} \quad x = \frac{2}{3}p + cp^{-2}$$

Putting (3) in (1), we get $y = \frac{1}{3}p^2 + 2cp^{-1}$ ---- (4)

(3) & (4) together constitute the general solution of (1).

11.16. EQUATIONS SOLVABLE FOR x

If the equation is solvable for x, we can express x explicitly in terms of y and p. Thus, the equations of this type can be put as $x = f(y, p)$ ...(1)

Differentiating (1) w.r.t. y, we get $\frac{dx}{dy} = \frac{1}{p} = F\left(y, p, \frac{dp}{dy}\right)$ ...(2)

Equation (2) is a differential equation of first order in p and y.

Suppose the solution of (2) is $\phi(y, p, c) = 0$ ...(3)

Now elimination of p from (1) and (3) gives the required solution.

If p cannot be easily eliminated, then we solve equations (1) and (3) for x and y to get

$$x = \phi_1(p, c),\ y = \phi_2(p, c)$$

These two relations together constitute the solution of the given equation with p as parameter.

Example 1. *Solve* $y = 2px + y^2p^3$. (Rewa, 1990)

Sol. Solving for x, we have $x = \frac{1}{2}\left(\frac{y}{p} - y^2p^2\right)$

Differentiating both sides w.r.t. y

$$\frac{dx}{dy} = \frac{1}{p} = \frac{1}{2}\left(\frac{1}{p} - \frac{y}{p^2}\cdot\frac{dp}{dy} - 2yp^2 - 2y^2p\frac{dp}{dy}\right)$$

or

$$2p = p - y\frac{dp}{dy} - 2yp^4 - 2y^2p^3\frac{dp}{dy}$$

or $$p + 2yp^4 + y\frac{dp}{dy} + 2y^2p^3\frac{dp}{dy} = 0 \quad \text{or} \quad p(1 + 2yp^3) + y\frac{dp}{dy}(1 + 2yp^3) = 0$$

or $$\left(p + y\frac{dp}{dy}\right)(1 + 2yp^3) = 0$$

Discarding the factor $(1 + 2yp^3)$, we have $p + y\dfrac{dp}{dy} = 0$ or $\dfrac{dy}{y} + \dfrac{dp}{p} = 0$

Integrating $\quad \log y + \log p = \log c \quad$ or $\quad py = c \quad$ or $\quad p = \dfrac{c}{y}$

Putting this value of p in the given equation, we have $y = \dfrac{2cx}{y} + \dfrac{c^3}{y}$ or $y^2 = 2cx + c^3$

which is the required solution.

Example 2. *Solve* $p = tan\left(x - \dfrac{p}{1 + p^2}\right)$.

Sol. Solving for x, we have $x = \tan^{-1} p + \dfrac{p}{1 + p^2}$ ...(1)

Differentiating both sides w.r.t. y, $\dfrac{dx}{dy} = \dfrac{1}{p} = \dfrac{1}{1 + p^2}\cdot\dfrac{dp}{dy} + \dfrac{(1 + p^2) - 2p^2}{(1 + p^2)^2}\cdot\dfrac{dp}{dy}$

or $$\frac{1}{p} = \frac{2(1 + p^2) - 2p^2}{(1 + p^2)^2}\frac{dp}{dy} \quad \text{or} \quad dy = \frac{2p}{(1 + p^2)^2}dp$$

Integrating $\quad y = c - \dfrac{1}{1 + p^2}$ ...(2)

Equations (1) and (2) together constitute the general solution.

11.17. CLAIRAUT'S EQUATION

An equation of the form $\quad y = px + f(p)$ ...(1)

is known as Clairaut's equation.

Differentiating (1) w.r.t. x, we get $p = p + x\dfrac{dp}{dx} + f'(p)\dfrac{dp}{dx}$ or $[x + f'(p)]\dfrac{dp}{dx} = 0$

Discarding the factor $[x + f'(p)]$, we have $\dfrac{dp}{dx} = 0$

Integrating $\quad p = c$

Putting $\quad p = c$ in (1), the required solution is $\overset{y}{\cancel{p}} = cx + f(c)$

Thus, the solution of Clairaut's equation is obtained by writing c for p.

Example 1. *Solve* $(y - px)(p - 1) = p$.

Sol. The given equation can be written as $y - px = \dfrac{p}{p-1}$ or $y = px + \dfrac{p}{p-1}$

This is of Clairaut's form. Hence putting c for p, the solution is $y = cx + \dfrac{c}{c-1}$.

Note. Many differential equations can be reduced to Clairaut's form by suitably changing the variables.

Example 2. *Solve* $e^{4x}(p - 1) + e^{2y} p^2 = 0$.

Sol. [In problems involving e^{lx} and e^{my}, put $X = e^{kx}$ and $Y = e^{ky}$, where k is the H.C.F. of l and m].

Put $\quad X = e^{2x}$ and $Y = e^{2y}$

so that $\quad dX = 2e^{2x}\,dx$ and $dY = 2e^{2y}\,dy$

$$\therefore \quad p = \frac{dy}{dx} = \frac{e^{2x}}{e^{2y}} \frac{dY}{dX} = \frac{X}{Y} P, \text{ where } P = \frac{dY}{dX}$$

The given equation becomes $X^2\left(\dfrac{X}{Y}P - 1\right) + Y \cdot \dfrac{X^2}{Y^2} P^2 = 0$

or $\quad XP - Y + P^2 = 0$ or $Y = PX + P^2$ which is of Clairaut's form

$\therefore$ Its solution is $Y = cX + c^2$ and hence $e^{2y} = ce^{2x} + c^2$.

5 **Example 3.** *Solve* $(px - y)(py + x) = 2p$. (Bangalore, 19

Sol. Put $\quad X = x^2$ and $Y = y^2$

so that $\quad dX = 2x\,dx$ and $dY = 2y\,dy$

$$\therefore \quad p = \frac{dy}{dx} = \frac{x}{y}\frac{dY}{dX} = \frac{\sqrt{X}}{\sqrt{Y}} P, \text{ where } P = \frac{dY}{dX}$$

The given equation becomes $\left(\dfrac{\sqrt{X}}{\sqrt{Y}} P \cdot \sqrt{X} - \sqrt{Y}\right)\left(\dfrac{\sqrt{X}}{\sqrt{Y}} P \cdot \sqrt{Y} + \sqrt{X}\right) = 2\dfrac{\sqrt{X}}{\sqrt{Y}} P$

or $\quad (PX - Y)(P + 1) = 2P$ or $PX - Y = \dfrac{2P}{P+1}$

or $\quad Y = PX - \dfrac{2P}{P+1}$ which is of Clairaut's form.

$\therefore$ Its solution is $Y = cX - \dfrac{2c}{c+1}$ and hence $y^2 = cx^2 - \dfrac{2c}{c+1}$.